ZCAS University

CCS2131 DATA STRUCTURES AND ALGORITHMS

SEMESTER FINAL EXAMINATION

WEDNESDAY 7 JUNE, 2023

16:30 - 19:30 HOURS

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. **Candidates must not turn this page until the invigilator tells them to do so. '**

SECTION A: Question 1 is compulsory and must be attempted

Question 1

a) Insertion sort is a simple sorting algorithm that works by repeatedly inserting elements into a sorted list. Using insertion sort, sort the numbers below.

[21, 33, 44, 55, 66, 22, 34, 45, 56, 67, 23, 35, 46, 57, 68, 24, 36, 47, 58, 69]

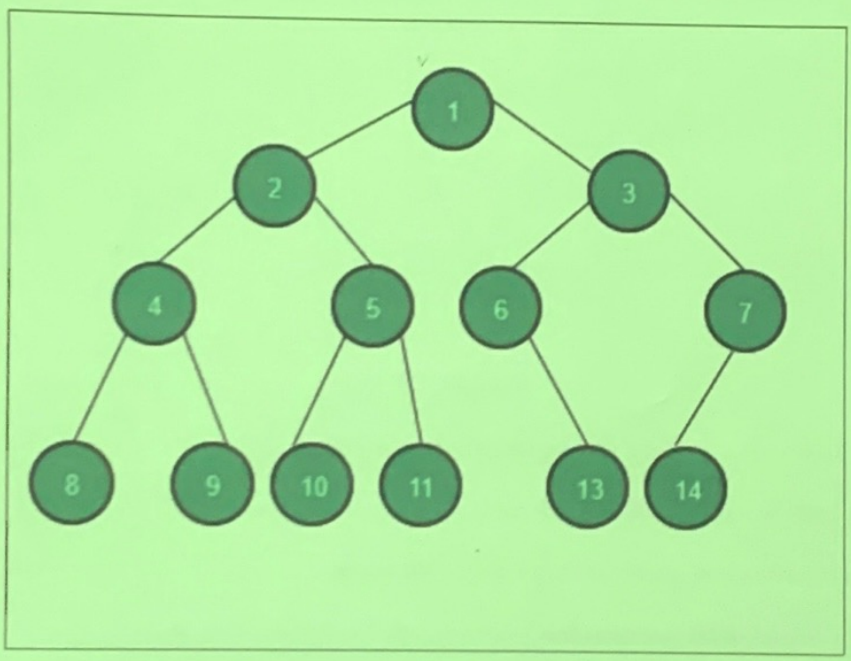
(10 Marks)

b) Given the following:

[23, 53, 34, 76, 45, 67, 88, 99, 20, 50, 30, 70, 40, 60, 80, 90, 21, 51, 31, 71, 41, 61]

Create a ***maximum heap tree***. Show all your workings. **(10 Marks)**

c) Refer to the binary tree below. Traverse the tree using ***preorder*** and ***postorder*** traversal. Show the final output sequence of traversals for each. **(10 Marks)**



pre root, left sub, right sub

post left sub, right sub, root

d) In-order traversal visits the left subtree of a node, then the node itself, and then the right subtree of the node. Write a C++ ***function*** that will handle ***inorder*** traversal.

(10 Marks)

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) A data structure is a way of organizing data so that it can be easily accessed and manipulated. List ***five*** common data structures used in programming.

(5 Marks)

b) A ***stack*** is a data structure that follows the last-in, first-out (LIFO) principle. This means that the last element that is added to the stack is the first element that is removed. With the aid of of an example, demonstrate three common operations on a stack. **(7 Marks)**

c) Write a C++ ***program*** to check if a given string is a palindrome. A palindrome is a word, phrase that reads the same backward as forward, such as madam or racecar. ***Hint***: use an array data structure. **(8 Marks)**

[Total: 20 Marks]

Question 3

a) Big O notation is a powerful tool for analyzing the efficiency of algorithms. Calculate the complexity of the code below.

```
int main() {
        int sum = 0;
        int num = 0
        while(num!=5){
                sum = sum+i;
                num= num+1;
        }
                cout << "Sum = " << sum;
                return 0;
}
```

(5 Marks)

b) With the aid of examples distinguish between binary trees and graph data structures **(5 Marks)**

c) Hashing is a technique used in computer science to convert data of arbitrary size into a fixed-size value, known as a hash code or hash value. Suppose we assign 1-26 to alphabetic characters a -z respectively, using a hash table, calculate the hash for the following text:

"Alice in wonderland

Assume the hash index starts at 0. **(10 Marks)**

[Total: 20 Marks]

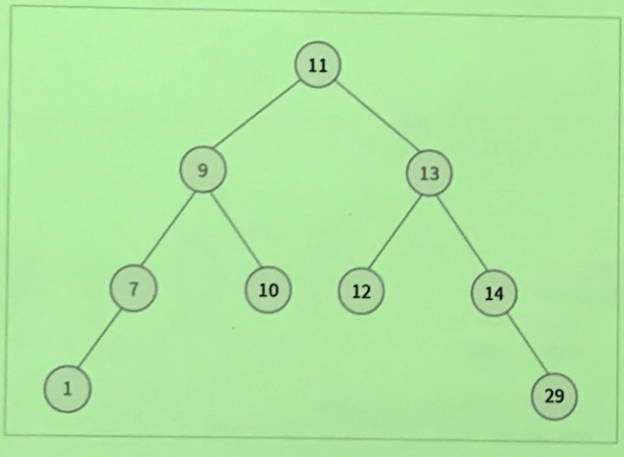
Question 4

a) Bubble sort is a simple sorting algorithm that repeatedly compares adjacent elements in an array and swaps them if they are in the wrong order. Bubble sort the numbers given below:

[21, 33, 44, 55, 66, 22, 34, 45, 56, 67, 23, 35, 46, 57, 68, 24, 36, 47, 58, 69]

(8 Marks)

b) A splay tree is a self-balancing binary search tree with the additional property that recently accessed elements are quick to access again. Given the tree below, splay the tree by searching for 10. **(7 Marks)**



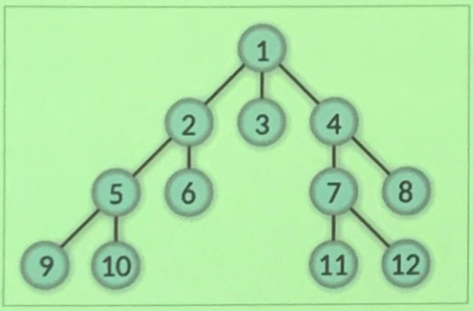
[Total: 20 Marks]

c) The Fibonacci sequence is a series of numbers where each number is the sum of the two preceding numbers. Write c++ ***function*** to calculate Fibonacci numbers **(5 Marks)**

Question 5

a) Analysis of algorithms is the process of finding the computational complexity of algorithms. Explain why it is important to analyse algorithms. **(4 Marks)**

b) Traverse the graph shown below using Breadth First Search (BFS) method. Show the final traversal out put **(8 Marks)**



c) In graph theory and computer science, an adjacency matrix is a square matrix used to represent a finite graph. Produce a graph from the adjacency matrix give below. **(8 Marks)**

	1	2	3	4
1	0	0	1	0
2	0	0	1	1
3	1	0	0	1
4	0	1	0	1

[Total: 20 Marks]

END OF EXAMINATION